

Infinite Series

Real Analysis

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Series

Absolute
ConvergenceTests for
Convergence

Series of Functions

Definition

If $X := (x_n)$ is a sequence in $\mathbb{R}$, then the **infinite series** (or simply the **series**) generated by X is the sequence $S := (s_k)$ defined by

$$s_1 := x_1, \quad s_2 := s_1 + x_2 (= x_1 + x_2),$$

$$\dots, \quad s_k := s_{k-1} + x_k (= x_1 + x_2 + \dots + x_k), \quad \dots$$

The numbers x_n are called the **terms of the series** and the numbers s_k are called the **partial sums of this series**. If $\lim S$ exists, we say that this series is **convergent** and call this limit the **sum** or the **value of this series**. If this limit does not exist, we say that the series S is **divergent**.

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Theorem

Let (x_n) be a sequence of nonnegative real numbers. Then the series $\sum x_n$ converges if and only if the sequence $S = (s_k)$ of partial sums is bounded. In this case,

$$\sum_{n=1}^{\infty} x_n = \lim(s_k) = \sup\{s_k : k \in \mathbb{N}\}.$$

Definition

Let $X := (x_n)$ be a sequence in $\mathbb{R}$. We say that the series $\sum x_n$ is **absolutely convergent** if the series $\sum |x_n|$ is convergent in $\mathbb{R}$. A series is said to be **conditionally (or nonabsolutely) convergent** if it is convergent, but it is not absolutely convergent.

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Theorem

If a series in $\mathbb{R}$ is absolutely convergent, then it is convergent.

Theorem

If a series $\sum x_n$ is convergent, then any series obtained from it by grouping the terms is also convergent and to the same value.

Definition

A series $\sum y_k$ in $\mathbb{R}$ is a **rearrangement of a series** $\sum x_n$ if there is a bijection f of $\mathbb{N}$ onto $\mathbb{N}$ such that $y_k = x_{f(k)}$ for all $k \in \mathbb{N}$.

Theorem (Rearrangement Theorem)

Let $\sum x_n$ be an absolutely convergent series in $\mathbb{R}$. Then any rearrangement $\sum y_k$ of $\sum x_n$ converges to the same value.

Theorem (Limit Comparison Test, II)

Suppose that $X := (x_n)$ and $Y := (y_n)$ are nonzero real sequences and suppose that the following limit exists in $\mathbb{R}$:

$$r := \lim \frac{x_n}{y_n}.$$

- (a) If $r \neq 0$, then $\sum x_n$ is absolutely convergent if and only if $\sum y_n$ is absolutely convergent.
- (b) If $r = 0$ and if $\sum y_n$ is absolutely convergent, then $\sum x_n$ is absolutely convergent.

Theorem (Root Test)

Let $X := (x_n)$ be a sequence in $\mathbb{R}$.

(a) If there exist $r \in \mathbb{R}$ with $r < 1$ and $K \in \mathbb{N}$ such that

$$|x_n|^{1/n} \leq r \quad \text{for } n \geq K,$$

then the series $\sum x_n$ is absolutely convergent.

(b) If there exists $K \in \mathbb{N}$ such that

$$|x_n|^{1/n} \geq 1 \quad \text{for } n \geq K,$$

then the series $\sum x_n$ is divergent.

Corollary

Let $X := (x_n)$ be a sequence in $\mathbb{R}$ and suppose that the limit

$$r := \lim |x_n|^{1/n}$$

exists in $\mathbb{R}$. Then $\sum x_n$ is absolutely convergent when $r < 1$ and is divergent when $r > 1$.

Theorem (Ratio Test)

Let $X := (x_n)$ be a sequence of nonzero real numbers.

(a) If there exist $r \in \mathbb{R}$ with $0 < r < 1$ and $K \in \mathbb{N}$ such that

$$\left| \frac{x_{n+1}}{x_n} \right| \leq r \quad \text{for } n \geq K,$$

then the series $\sum x_n$ is absolutely convergent.

(b) If there exists $K \in \mathbb{N}$ such that

$$\left| \frac{x_{n+1}}{x_n} \right| \geq 1 \quad \text{for } n \geq K,$$

then the series $\sum x_n$ is divergent.

Corollary

Let $X := (x_n)$ be a nonzero sequence in $\mathbb{R}$ and suppose that the limit

$$r := \lim \frac{x_{n+1}}{x_n}$$

exists in $\mathbb{R}$. Then $\sum x_n$ is absolutely convergent when $r < 1$ and is divergent when $r > 1$.

Theorem (Integral Test)

Let f be a positive, decreasing function on $\{t : t \geq 1\}$. Then the series $\sum_{k=1}^{\infty} f(k)$ converges if and only if the improper integral

$$\int_1^{\infty} f(t) dt = \lim_{b \rightarrow \infty} \int_1^b f(t) dt$$

exists. In the case of convergence, the partial sum $s_n = \sum_{k=1}^n f(k)$ and the sum

$s = \sum_{k=1}^{\infty} f(k)$ satisfy the estimate

$$\int_{n+1}^{\infty} f(t) dt \leq s - s_n \leq \int_n^{\infty} f(t) dt.$$

Theorem (Raabe's Test)

Let $X := (x_n)$ be a sequence of nonzero real numbers.

(a) If there exist numbers $a > 1$ and $K \in \mathbb{N}$ such that

$$\left| \frac{x_{n+1}}{x_n} \right| \leq 1 - \frac{a}{n} \quad \text{for } n \geq K,$$

then $\sum x_n$ is absolutely convergent.

(b) If there exist real numbers $a \leq 1$ and $K \in \mathbb{N}$ such that

$$\left| \frac{x_{n+1}}{x_n} \right| \geq 1 - \frac{a}{n} \quad \text{for } n \geq K,$$

then $\sum x_n$ is not absolutely convergent.

Corollary

Let $X := (x_n)$ be a nonzero sequence in $\mathbb{R}$ and let

$$a := \lim n \left(1 - \frac{x_{n+1}}{x_n} \right),$$

whenever this limit exists. Then $\sum x_n$ is absolutely convergent when $a > 1$ and is not absolutely convergent when $a < 1$.

Definition

A sequence $X := (x_n)$ of nonzero real numbers is said to be **alternating** if the terms $(-1)^{n+1}x_n$, $n \in \mathbb{N}$, are all positive (or all negative) real numbers. If the sequence $X = (x_n)$ is alternating, we say that the series $\sum x_n$ it generates is an **alternating series**.

Theorem (Alternating Series Test)

Let $Z := (z_n)$ be a decreasing sequence of strictly positive numbers with $\lim(z_n) = 0$. Then the alternating series $\sum (-1)^{n+1} z_n$ is convergent.

Theorem (Abel's Lemma)

Let $X := (x_n)$ and $Y := (y_n)$ be sequences in $\mathbb{R}$ and let the partial sums of $\sum y_n$ be denoted by (s_n) with $s_0 := 0$. If $m > n$, then

$$\sum_{k=n+1}^m x_k y_k = x_m s_m - x_{n+1} s_n + \sum_{k=n+1}^{m-1} (x_k - x_{k+1}) s_k.$$

Theorem (Dirichlet's Test)

If $X := (x_n)$ is a decreasing sequence with $\lim x_n = 0$, and if the partial sums (s_n) of $\sum y_n$ are bounded, then the series $\sum x_n y_n$ is convergent.

Theorem (Abel's Test)

If $X := (x_n)$ is a convergent monotone sequence and the series $\sum y_n$ is convergent, then the series $\sum x_n y_n$ is also convergent.

Definition

If (f_n) is a sequence of functions defined on a subset D of $\mathbb{R}$ with values in $\mathbb{R}$, the sequence of partial sums (s_n) of the infinite series $\sum f_n$ is defined for x in D by

$$s_1(x) := f_1(x), \quad s_2(x) := s_1(x) + f_2(x), \quad \dots, \quad s_{n+1}(x) := s_n(x) + f_{n+1}(x), \quad \dots$$

In case the sequence (s_n) of functions converges on D to a function f , we say that the infinite series of functions $\sum f_n$ **converges to f on D** . If the series $\sum |f_n(x)|$ converges for each x in D , we say that $\sum f_n$ is **absolutely convergent on D** . If the sequence (s_n) of partial sums is uniformly convergent on D to f , we say that $\sum f_n$ is **uniformly convergent on D** , or that it converges to f uniformly on D .

Theorem

If f_n is continuous on $D \subseteq \mathbb{R}$ to $\mathbb{R}$ for each $n \in \mathbb{N}$ and if $\sum f_n$ converges to f uniformly on D , then f is continuous on D .

Theorem

Suppose that the real-valued functions f_n , $n \in \mathbb{N}$, are Riemann integrable on the interval $J := [a, b]$. If the series $\sum f_n$ converges to f uniformly on J , then f is Riemann integrable and

$$\int_a^b f = \sum_{n=1}^{\infty} \int_a^b f_n.$$

Theorem

For each $n \in \mathbb{N}$, let f_n be a real-valued function on $J := [a, b]$ that has a derivative f'_n on J . Suppose that the series $\sum f_n$ converges for at least one point of J and that the series of derivatives $\sum f'_n$ converges uniformly on J . Then there exists a real-valued function f on J such that $\sum f_n$ converges uniformly on J to f . In addition, f has a derivative on J and $f' = \sum f'_n$.

Theorem (Cauchy Criterion)

Let (f_n) be a sequence of functions on $D \subseteq \mathbb{R}$ to $\mathbb{R}$. The series $\sum f_n$ is uniformly convergent on D if and only if for every $\varepsilon > 0$ there exists an $M(\varepsilon)$ such that if $m > n \geq M(\varepsilon)$, then

$$|f_{n+1}(x) + \cdots + f_m(x)| < \varepsilon \quad \text{for all } x \in D.$$

Theorem (Weierstrass M-Test)

Let (M_n) be a sequence of positive real numbers such that $|f_n(x)| \leq M_n$ for $x \in D$, $n \in \mathbb{N}$. If the series $\sum M_n$ is convergent, then $\sum f_n$ is uniformly convergent on D .

Definition

A series of real functions $\sum f_n$ is said to be a **power series around** $x = c$ if the function f_n has the form

$$f_n(x) = a_n(x - c)^n,$$

where a_n and c belong to $\mathbb{R}$ and where $n = 0, 1, 2, \dots$

Definition

Let $\sum a_n x^n$ be a power series. If the sequence $(|a_n|^{1/n})$ is bounded, we set $r := \limsup(|a_n|^{1/n})$; if this sequence is not bounded we set $r = +\infty$. We define the **radius of convergence** of $\sum a_n x^n$ to be given by

$$R := \begin{cases} 0 & \text{if } r = +\infty, \\ 1/r & \text{if } 0 < r < +\infty, \\ +\infty & \text{if } r = 0. \end{cases}$$

The **interval of convergence** is the open interval $(-R, R)$.

Theorem (Cauchy-Hadamard Theorem)

If R is the radius of convergence of the power series $\sum a_n x^n$, then the series is absolutely convergent if $|x| < R$ and is divergent if $|x| > R$.

Theorem

Let R be the radius of convergence of $\sum a_n x^n$ and let K be a closed and bounded interval contained in the interval of convergence $(-R, R)$. Then the power series converges uniformly on K .

Theorem

The limit of a power series is continuous on the interval of convergence. A power series can be integrated term-by-term over any closed and bounded interval contained in the interval of convergence.

Theorem (Differentiation Theorem)

A power series can be differentiated term-by-term within the interval of convergence. In fact, if

$$f(x) = \sum_{n=0}^{\infty} a_n x^n, \quad \text{then} \quad f'(x) = \sum_{n=1}^{\infty} n a_n x^{n-1} \quad \text{for} \quad |x| < R.$$

Both series have the same radius of convergence.

Theorem (Uniqueness Theorem)

If $\sum a_n x^n$ and $\sum b_n x^n$ converge on some interval $(-r, r)$, $r > 0$, to the same function f , then

$$a_n = b_n \quad \text{for all } n \in \mathbb{N}.$$

Remark

*If a function f has derivatives of all orders at a point c in $\mathbb{R}$, then we calculate the **Taylor coefficients** by $a_k := f^{(k)}(c)/k!$ for $k \in \mathbb{N}$ and in this way obtain a power series with these coefficients. However, it is not necessarily true that the resulting power series converges to the function f in an interval about c . The issue of convergence is resolved by the remainder term R_n in Taylor's theorem with the remainder.*

Definition

Let f be a function with derivatives of all orders at a point $c \in \mathbb{R}$. We will write

$$f(x) = \sum_{k=0}^{\infty} \frac{f^{(k)}(c)}{k!} (x - c)^k \quad \text{for } |x - c| < R$$

if and only if the sequence $(R_n(x))$ of remainders (defined as in Taylor's theorem with the remainder) converges to 0 for each x in some neighbourhood $V_R(c)$. In this case we say that the power series is the **Taylor expansion** of f at c and the partial sums

$$P_n(x) = \sum_{k=0}^n \frac{f^{(k)}(c)}{k!} (x - c)^k$$

are called the **Taylor polynomials** for f .